

Section A

Answer **all** the questions in this section.

1. Lipid vesicles (i.e. spherical lipid bilayers) that contain $\text{Na}^+\text{-K}^+$ pumps as the sole membrane protein were prepared. For the sake of simplicity, assume that each pump transports one Na^+ one way and one K^+ the other way in each pumping cycle as shown in **Fig. 1.1**. Each $\text{Na}^+\text{-K}^+$ pump is oriented so that the portion of the molecule that normally faces the cytosol faces the outside of the vesicle.

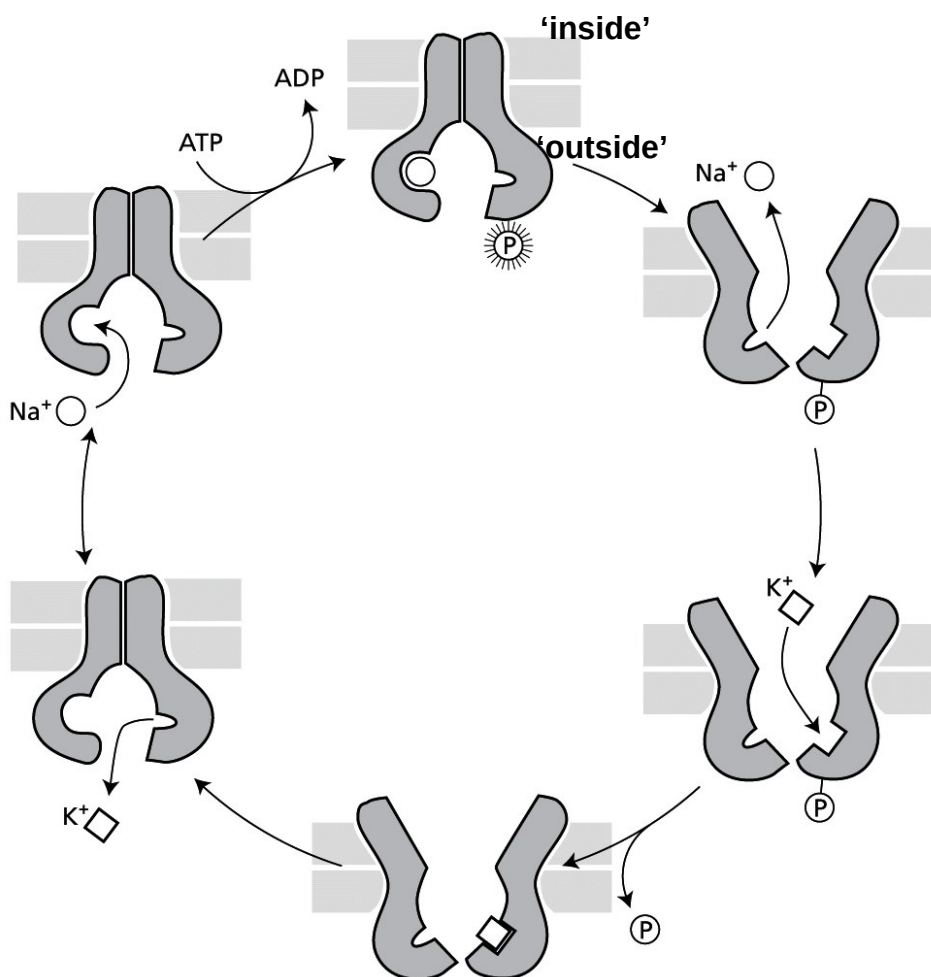


Fig. 1.1

With reference to **Fig. 1.1**, explain what would happen under each of the following conditions.

- (a) The solution inside and outside the vesicles contains both Na^+ and K^+ ions but no ATP.

[1]

1. no ions will be pumped in the absence of ATP;;

1. (b) The solution inside the vesicles contains both Na^+ and K^+ ions; the solution outside contains both ions, as well as ATP. [2]
1. The pumps will hydrolyse ATP;
 2. for the transport of Na^+ into the vesicles and K^+ out of the vesicles;
 3. This generates oppositely oriented concentration gradients;
 4. of Na^+ and K^+ across the membrane;
 5. The pumps will stop; when the ATP runs out;
- (c) The solution inside contains Na^+ ; the solution outside contains Na^+ and ATP. [3]
1. The pump will bind Na^+ ; hydrolyse ATP and transfer P_i to itself; and transport Na^+ against concentration gradient;
 2. The pump will then stop; because (i) K^+ is absent; and (ii) the conformational changes are strictly sequential;
- (d) Describe how the Na^+/K^+ pumps are responsible for maintenance of the resting membrane potential in a motor neurone. [2]
1. The pump will move 3 Na^+ out of the cell and 2 K^+ into the cell; against their respective concentration gradients;
 2. There is a net loss of 1 positive charge; to the extracellular fluid;
 3. Hence, the inside of the cell has a more negative voltage compared to the outside;

[Total: 8]

2.

PART I

In muscle tissue, the conversion of glucose to lactic acid during anaerobic respiration can be represented using a four-stage process as shown in **Fig. 2.1**.

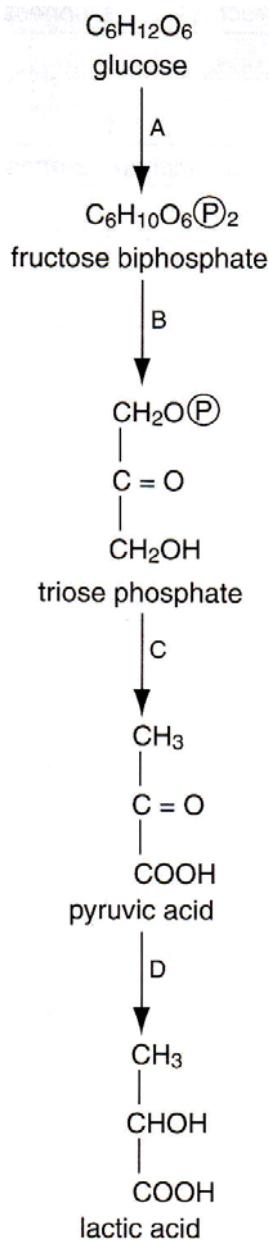


Fig. 2.1

(a) Describe stages A, B, C and D.

[4]

A: phosphorylation of; glucose to fructose-1,6-bisphosphate;

B: splitting of; fructose-1,6-bisphosphate to 2 molecules of triose phosphate;

C: oxidation of; TP;
substrate level phosphorylation of; ADP;

Reject: dephosphorylation of TP

Warning: oxidation of respiratory substrates and SLP are separate concepts!!!

D: reduction of; pyruvate to lactate;

2. (b) Is there a net change in the concentrations of NAD^+ and NADPH/H^+ in the whole process? Explain your answer. [2]
1. no net change;
 2. NAD^+ accepts hydrogen atoms from triose phosphate (i.e. oxidises triose phosphate); and forms NADH/H^+ ;
 3. NADH/H^+ is then used to reduce pyruvate to lactate; and NAD^+ is regenerated;
- (c) If one glucose molecule is converted to two lactic acid molecules, what is the net synthesis of ATP? Explain your answer. [2]
1. net synthesis of 2 ATP molecules;
 2. 2 ATP molecules used to phosphorylate glucose / during energy investment phase;
 3. 4 ATP molecules synthesised by substrate level phosphorylation / during energy payoff phase;
 - Reject: 4 ATP molecules synthesised by oxidation of TP
 4. lactate fermentation does not yield ATP molecules;
- (d) What is the metabolic significance to the muscle of this net synthesis of ATP? [1]
1. net synthesis of ATP molecules enable muscle contraction to continue;
 2. even when no oxygen is available and oxidative phosphorylation does not occur;
 - Reject: increases the energy available to the muscle cells because anaerobic respiration produces only 1/19 the amount of ATP of aerobic respiration

PART II

Phosphofructokinase (PFK) has been isolated from the hyperthermophilic archaeon *Pyrococcus furiosus*. It was subjected to standard analysis to determine its basic catalytic properties. The processes under study were of the form

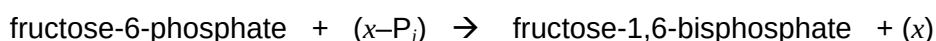


Fig. 2.2 demonstrates the increase in PFK activity with increasing ADP (i.e. $x-\text{P}_i$) concentration, i.e. increasing substrate concentration. The increase in PFK activity is determined by the increase in fructose-1,6-bisphosphate.

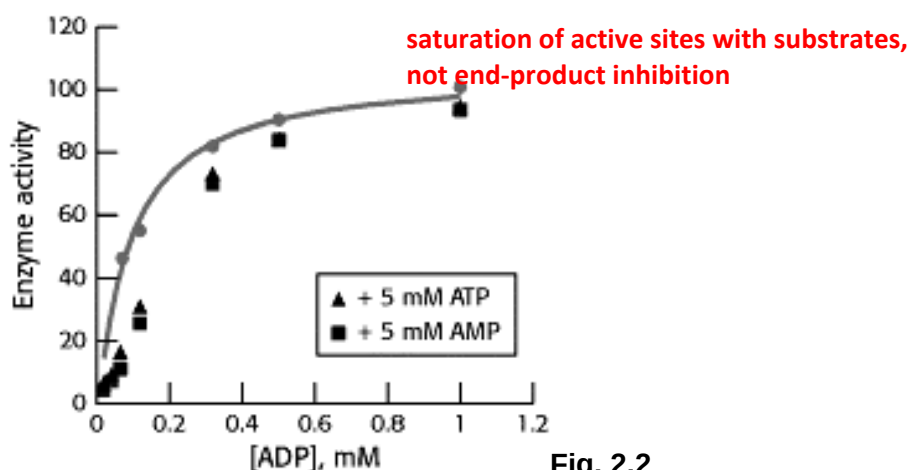


Fig. 2.2

2. (e) With reference to the information given, explain how *P. furiosus* PFK differs in action from normal PFK. [2]
1. source of the phosphate group;
 2. used to phosphorylate fructose-6-phosphate;
 3. is ADP;
 4. instead of ATP;
- Accept: normal PFK inhibited by high [ATP];; but *P. furiosus* PFK inhibited by high [AMP] or [ATP];;
- (f) With reference to **Fig. 2.2**, explain the effect(s) that AMP and ATP have on the reaction with ADP. [3]
1. both AMP and ATP have shapes similar to that of ADP;;
 2. both AMP and ATP behave as competitive inhibitors of PFK;;
 3. AMP or ATP prevents binding of ADP with the PFK active site / reduces the probability of effective collision between ADP and PFK;;
- Reject: reference to AMP and/or ATP as non-competitive / allosteric inhibitor

[Total: 14]

3.

PART I

Fig. 3.1 is a diagrammatic representation of the process of transcription.

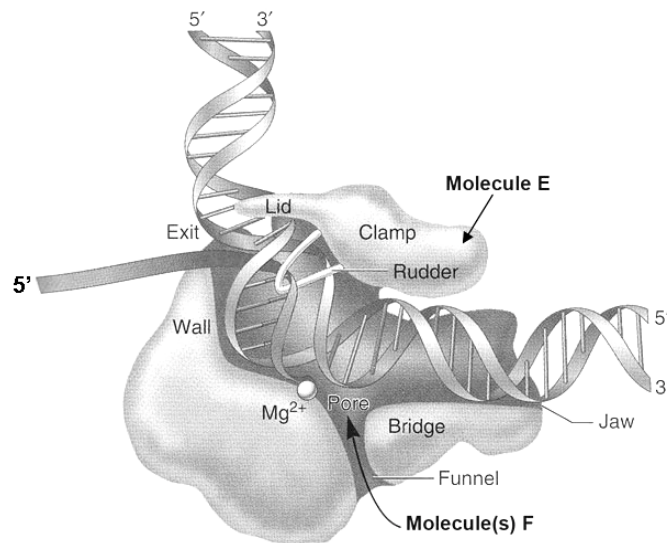


Fig. 3.1

- (a) Describe how the structure of molecule E is adapted to its role in transcription. [2]

Note: molecule E is RNA polymerase, not transcription factor!!!

1. molecule E has a specific active site;
Accept: active site being complementary to DNA
2. contact amino acid residues; that binds substrates such as DNA template, ribonucleotides; and nascent products such as the growing RNA strand;
3. catalytic amino acid residues; that catalyses phosphoester bond formation between growing RNA strand and incoming free nucleotide;
4. funnel allows entry of nucleoside triphosphates;
5. rudder separates the RNA-DNA hybrid;

- (b) Describe two ways by which the structure of molecule(s) F, which is / are adapted for transcription, differs from that of amino acids, which are adapted for translation. [2]

1. a molecule of F (dNTP) is made of a central deoxyribose attached to phosphate group and nitrogenous bases; but an amino acid is made of a central α carbon attached to amino group and carboxyl group;
2. identity of molecule F is determined by identity of nitrogenous base; but identity of amino acid is determined by identity of R-group;

- (c) Using evidence from Fig. 3.1, explain why the process of transcription is conservative. [2]

1. DNA double helix unwind and separate; to reveal the template strand;
2. after which both DNA strands rewind; to restore the double helix structure;

Note: conserve the DNA double helix

Reject: lengthy discourse on complementary base-pairing

3.

PART II

Double-stranded DNA of the human insulin gene (Note: it's a single, not many, genes!) is denatured and allowed to hybridise with mature mRNA isolated from pancreatic cells. Electron microscopy revealed heteroduplex molecules as shown in Fig. 3.2.

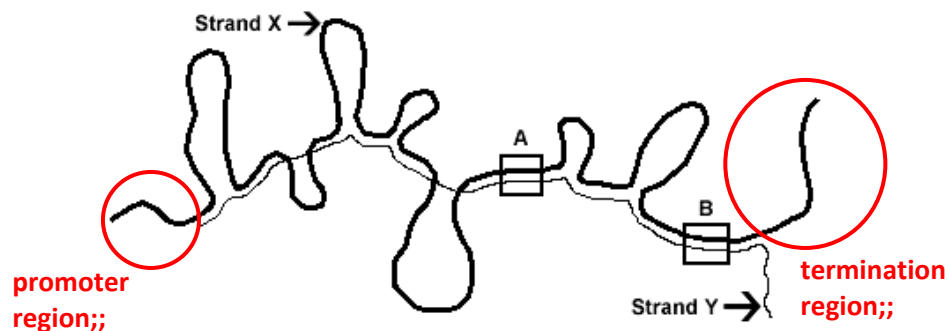


Fig. 3.2

- (d) (i) State the identity of strand X. Note: not identity of strand Y. [1]

template / transcribed / sense strand;; DNA

Reject: nonsense strand, double-stranded DNA because DNA had been denatured

- (ii) Using evidence from Fig. 3.2, explain the identity of strand X. [3]

1. strand X has a greater number of bases than strand Y;

Accept: strand X is longer than strand Y

Reject: strand X is thicker than strand Y because DNA is double-stranded

2. explain length as being due to the promoter; termination region; introns; that are not hybridised / paired with strand Y (i.e. mRNA) via complementary base pairing;

Reject: not joined to strand Y as "join" is too vague, intron loops being due to intra-strand base pairing.

3. explain template identity as being due to that fact that strand X is complementary to (i.e. hybridised with) strand Y (i.e. mRNA); at regions such as the coding regions; and UTRs; and thus provides the code / template for mRNA;

3. (e) Circle the locations of the promoter as well as the termination regions of the insulin gene on Fig. 3.2. Label the regions accordingly. [2]

- (f) What does the unpaired segment on strand Y represent? Give a reason for your answer. [2]

1. poly (A) tail;;

2. it is not coded for by the template DNA; and thus cannot form complementary base pairs with the template;

Accept: does not bind to strand X via complementary base pairing

Reject: because strand Y is unpaired because you need to paraphrase!!!

3. it is added to strand Y post-transcription;

[Total: 14]

4.

PART I

Fig. 4.1 is an electron micrograph showing a virus infecting a host cell.

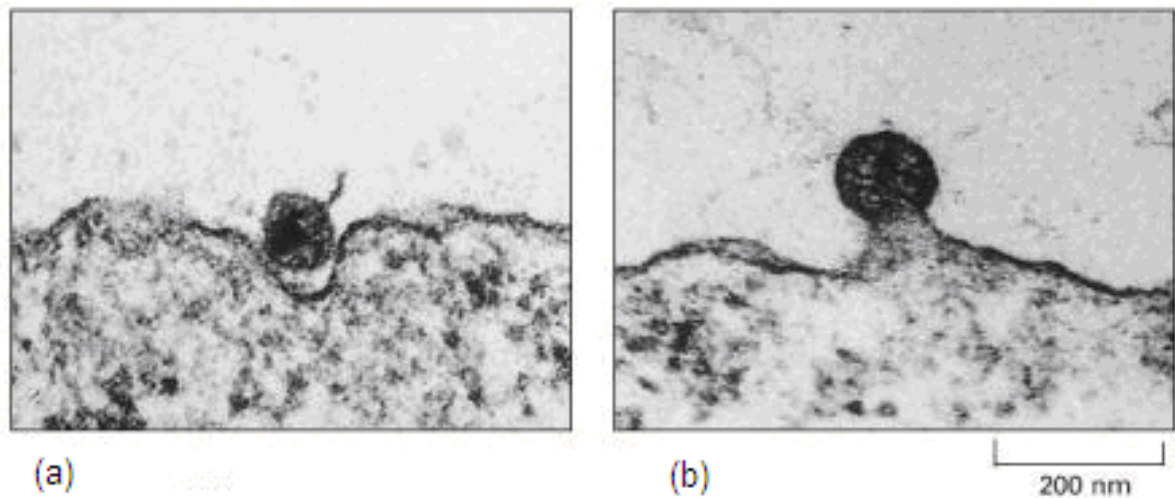


Fig. 4.1

- (a) Describe how an enveloped virus such as HIV enters a host cell. [3]
1. Glycoprotein 120 (gp120) on HIV envelope binds; to CD4 receptor; and chemokine receptors (e.g. CCR5 or CXCR4) on the plasma membrane of host cells;
 2. Once HIV binds, the viral envelope fuses with the plasma membrane of the host cell;;
 3. Capsid and RNA genome are released into the cytoplasm;
- (b) Suggest why the HIV causes damage only to some types of human cell. [1]
- Surface glycoproteins of virus can only bind / attach to complementary shape/specific receptors proteins in the cell membrane of host cell;;

4.

PART II

Several studies have already revealed that some CCR5 receptors belonging to the G protein-coupled receptors family might be necessary for the entry of the virus into the cell.

Fig. 4.2 is a 3-dimensional model of the human CCR5 receptor, which is a protein.

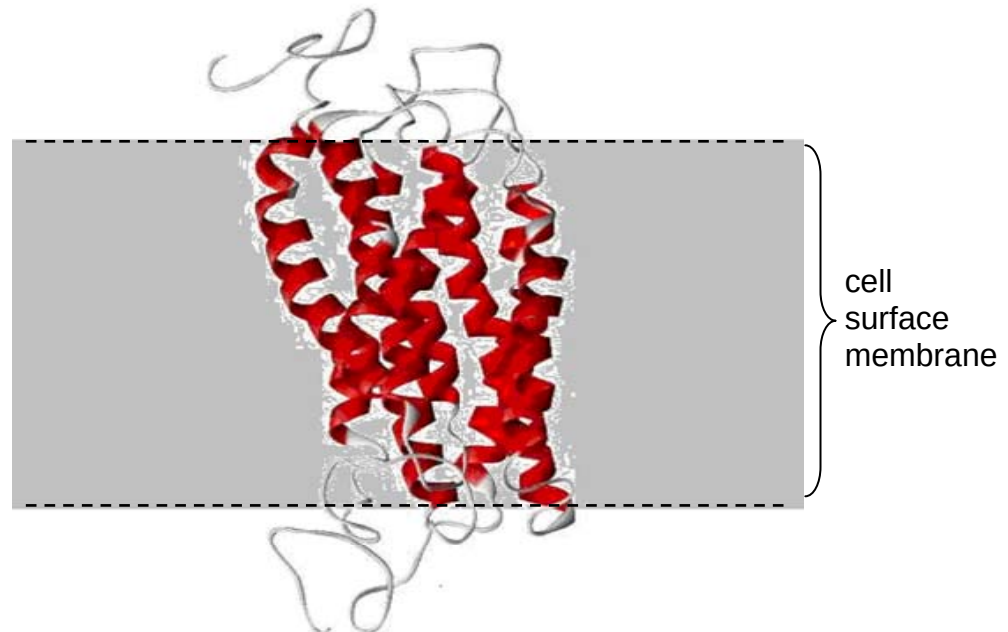


Fig. 4.2

(c) With reference to **Fig. 4.2**, describe how the polypeptide chain of CCR5 allows it to be maintained in the membrane. [4]

1. the polypeptide chain consists of both hydrophobic and hydrophilic amino acid residues;;
2. the primary structure folds into secondary structure / α -helix; then fold on its own to form the tertiary structure; maintained by tertiary bonds;
3. hydrophobic regions of the polypeptide interact via hydrophobic interactions with the hydrophobic hydrocarbon tails of the phospholipids;;
4. hydrophilic portions interact with the phosphate groups of the phospholipids; and the aqueous environment on either side of the membrane;

4. It was observed that a small percentage of high-risk populations showed either resistance or delayed development of the disease. The resistance of these rare individuals to HIV infection has now been explained by the discovery that they are homozygous for an allelic, nonfunctional variant of CCR5 caused by a 32-base-pair deletion from the coding region in the gene that codes for the CCR5 receptor.

(d) Explain **one** possible consequence of how the mutation results in resistance to the disease. [3]

1. change in DNA nucleotide sequence is transmitted to mRNA sequence during transcription and changes the reading frame downstream of the deletion;
 2. mRNA read in a series of triplets / codons on mRNA / codons are also read in a non-overlapping manner;
 3. result in change in amino acid sequence of the polypeptide;
- or
1. results in a stop codon;
 2. translation will be terminated prematurely;
 3. the resulting polypeptide will be truncated;
-
4. change in tertiary structure of CCR5;
 5. no longer complementary shape to virus glycoprotein;
 6. Virus cannot bind to CCR5, hence, cannot enter the cell;

[Total: 11]

5. The colour of the spines on the stems of raspberry plants are controlled by two genes, **A/a** and **B/b**. The genes are on different pairs of chromosomes.

Allele **A** produces a pink anthocyanin pigment in the spines. Allele **B** has no effect by itself, but increases the colour produced by allele **A** to give red spines. Alleles **a** and **b** have no effect on spine colour. In the absence of anthocyanin, the spines are green.

(a) Suggest how allele **B** may alter the expression of allele **A**. [2]

1. epistasis;
2. allele B increased expression/ transcription of allele A;
3. make more pigment/ enzyme to alter pigment/ change structure of pigment / AVP;;

Plants with the genotypes **AaBb** and **aabb** were cross-pollinated. The resulting seeds were sown and the seedlings grown until their stems developed spines.

(b) (i) Draw a genetic diagram of this cross to show the ratio of different phenotypes expected in the offspring. [4]

<i>Parental phenotypes</i>	red spines	x	green spines	;
<i>Parental genotypes</i>	AaBb	x	aabb	
<i>Gametes</i>	AB Ab aB ab	x	ab	::

<i>Punnett square</i>	AB	Ab	aB	ab
ab	AaBb red spines	Aabb pink spines	aaBb green spines	aabb green spines

<i>Offspring phenotypic ratio:</i>	1:	1:	2 ;
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offspring genotypes;; minus ½ for each mistakes
phenotypes related to genotypes;; A: key

5. (b) (ii) In any genetic cross, the observed results are usually different from the expected results.

Suggest *two* reasons why such a difference may occur, referring only to events after meiosis. [2]

1. sample too small;;
 2. chance variation/ variation statistically insignificant;;
 3. differential survival of sperm/ ova (with particular genotypes);;
 4. differential survival rate of zygotes (with particular genotypes);;
- any two*

- (iii) Explain what differences in the phenotypic ratio would be expected if genes **A/a** and **B/b** were on the same homologous pair of chromosomes, as shown in **Fig. 5.1**.

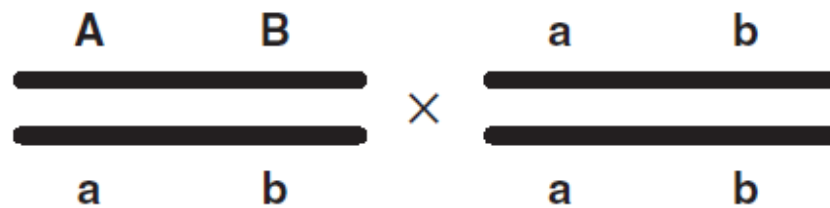


Fig. 5.1

[4]

1. ref more parental types;
2. i.e. many AaBb and aabb;
3. ref 1 : 1 ratio of these;
4. ref linkage;
5. ref fewer recombinants;
6. i.e. few Aabb and aaBb;
7. ref 1 : 1 ratio of these;
8. ref due to crossing over;
9. many red and green spined;
10. few / no, pink spined;
11. ref proportions depend on how close, loci / genes, are;

[Total: 12]

6. In Lake Tanganyika in Africa, there are six species of fish of the genus *Tropheus* and a much larger number of distinctly coloured subspecies of each of the six species. *Tropheus* species are small fish that are confined to isolated rocky habitats around the shores of Lake Tanganyika.

The six species evolved during the primary radiation phase when the lake was first filled, about 1.25 million years ago. They arose from river dwelling ancestors and then filled all available niches in the lake.

Secondary radiations into the many subspecies occurred during the last 200 000 years. Sometime during this period, the water level in the lake fell, resulting in the formation of three separate lake basins. These basins persisted for many thousands of years before the water level rose again.

Fig. 6.1 shows an outline map of the lake and the location of the three temporary basins caused by lowering of lake levels.

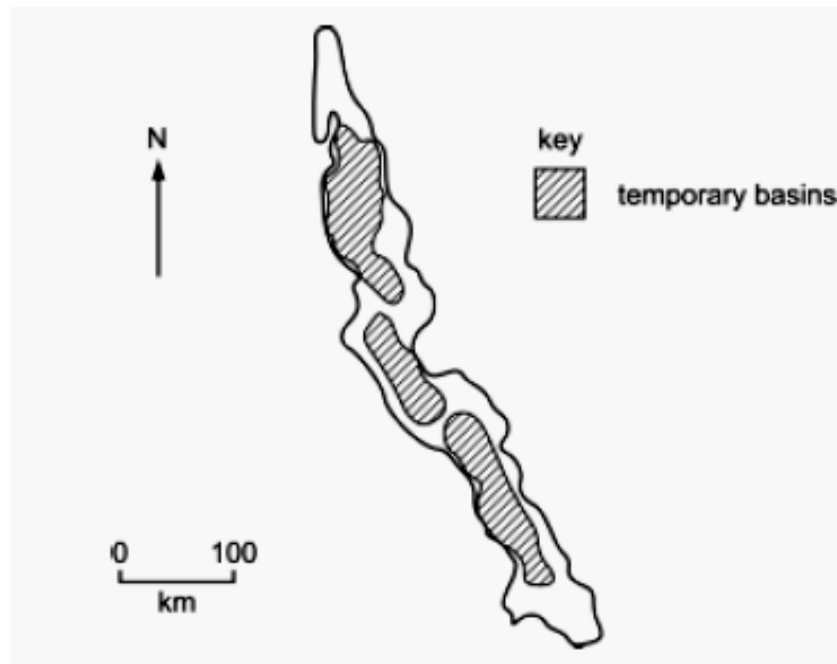


Fig. 6.1

- (a) (i) Explain how natural selection could have caused the evolution of the six closely related species in the primary radiation. [3]
1. variations in the population; due to mutation/ different alleles;
 2. different parts of the lake exerts different selection pressure;
 3. favouring different phenotypes;
 4. organisms with best-suited/favourable phenotype, survive and reproduce;
 5. passing on their genes/alleles to the next generation;
 6. reproductive/ behavioural/ sympatric/ allopatric/ geographical isolation/ accept hundreds of km apart/ no gene flow between different populations;
6. (a) (ii) Suggest how the lowering of the water level in the lake to form three separate lake basins could have caused the evolution of so many subspecies. [2]
1. ref. geographical barriers causing isolation;;
 2. preventing gene flow/interbreeding;;
 3. ref. different environments/habitats → different selection pressures;;

Recent research has compared DNA sequence of cytochrome b gene of the six different species of *Tropheus* (**Fig. 6.2**).

<i>T. brichardi</i>	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
<i>T. kasabae</i>	CTTTTAACTATGATAATAGCTTTTGTGGGCTACG
<i>T. annectens</i>	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
<i>T. moorii</i>	CTTTTAACTATGATAACAGCTTTTGTGGGTTACG
<i>T. polli</i>	CTTTTAACTATAATAACAGCTTTTGTGGGCTACG
<i>T. duboisi</i>	CTTTTAACCATGATAACGGCTTTTGTAGGCTATG
	***** ** ***** ***** ** ** *

Fig. 6.2 shows part of the cytochrome b DNA sequence

A family tree based on differences between the DNA sequences is shown in **Fig. 6.3**.

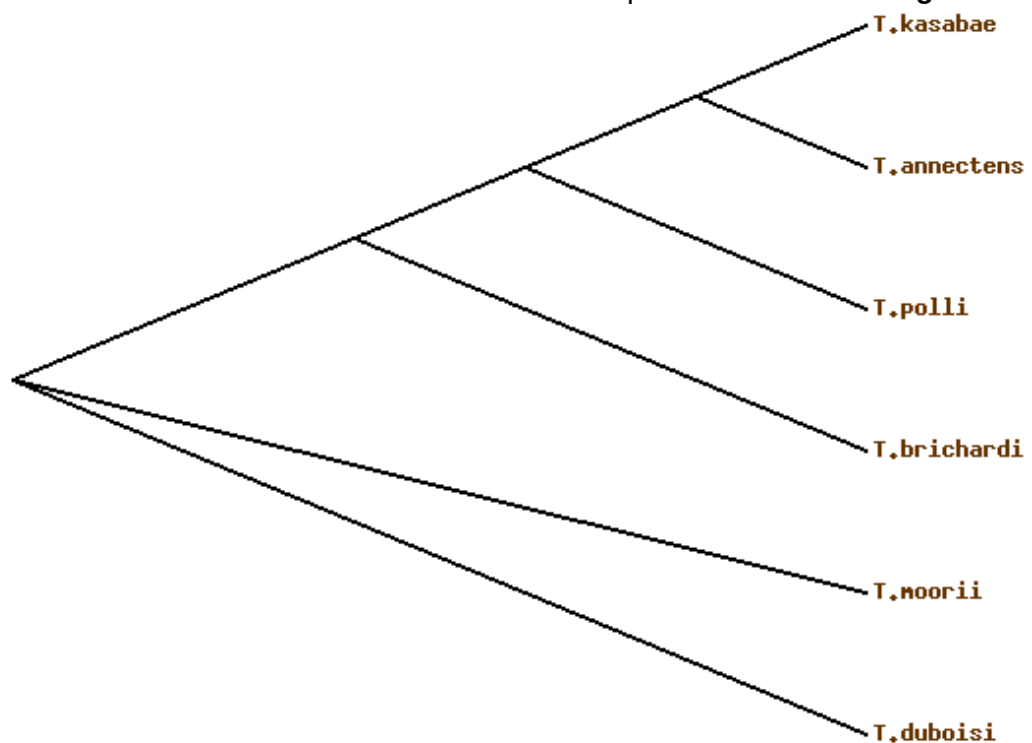


Fig. 6.3

6. (b) With reference to **Fig. 6.2**, explain why different species of *Tropheus* can have similar DNA sequence in their cytochrome b gene. [2]
1. they might have evolved from a common ancestor;;
 2. code for important protein involved respiration;;
 3. any mutation will result in non-functional protein which will be fatal to the organism;;
- (c) With reference to **Fig. 6.2 and 6.3**,
- (i) describe how the time of divergence of the species could be estimated. [2]
1. calibrate the molecular clock;
 2. compare the difference in the DNA sequence between species;
 3. determine the age of fossil record by dating;
 4. calculate the rate of mutation;
 5. no. of mutation is used to calculate the length of time since divergence;
- (ii) explain the advantage of the use of DNA sequence over amino acid sequence in determining the phylogeny of *Tropheus* species. [2]
1. change in DNA nucleotide sequence, does not change the type of amino acid;;
 2. due to degeneracy of genetic code;;

[Total: 11]

7. Diabetes mellitus is often due to the destruction of the β -cells in the islets of Langerhans in the pancreas. **Fig. 7.1** shows the changes in blood glucose concentration over a day in which an untreated diabetic person and a non-diabetic person take the same meals and exercise. Blood glucose was measured in both individuals every hour. At mealtimes, the measurements were made just before the meal.

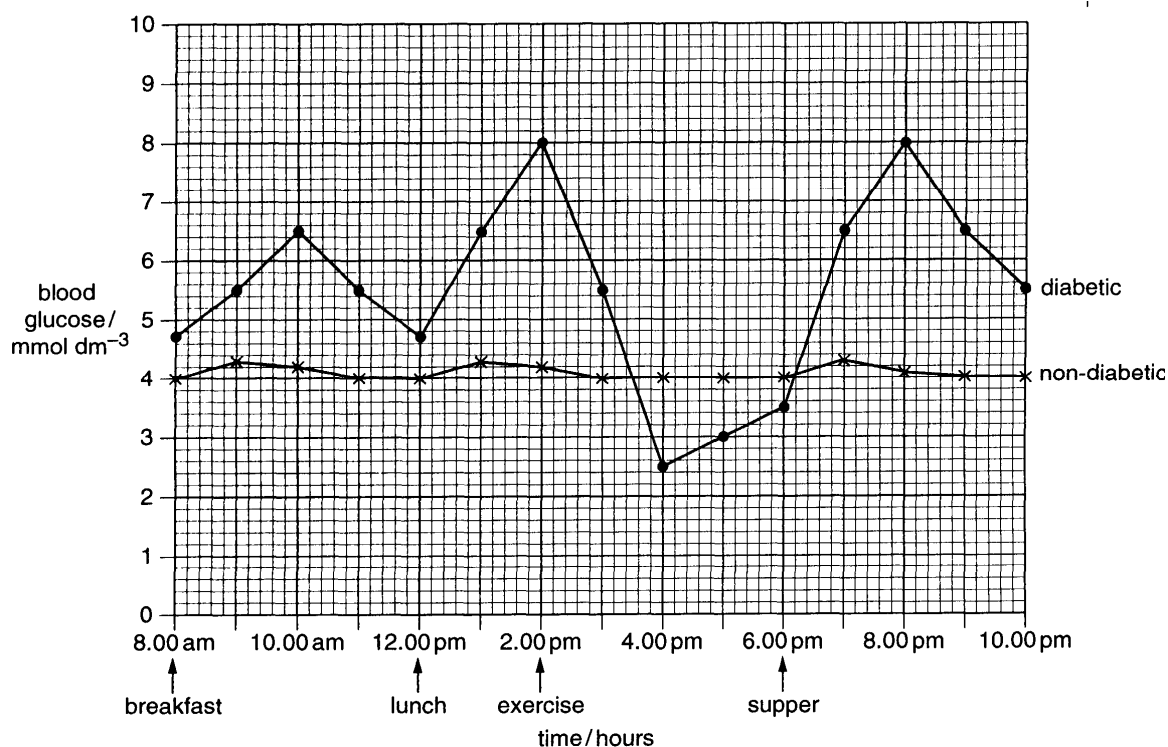


Fig. 7.1

- (a) Account for the differences in blood glucose at 10.00 am. [3]

1. reference to glucose being absorbed into the bloodstream;
2. diabetic produces little / no insulin;
3. blood glucose concentration of diabetic rises to 6.4 mmol dm^{-3} ;
4. normal person produces insulin;
5. glucose removed and taken up by liver / muscle cells and stored as glycogen;
6. blood glucose concentration of normal person remained constant at 4.2 mmol dm^{-3} ;

- (b) Explain why, following exercise at 2.00 pm,

- (i) the blood glucose in the diabetic person dropped; [2]
1. exercise increases need for ATP / respiration;;
 2. muscles take up glucose from the blood;
 3. in diabetic little or no glycogen stored;

Note:

- a. **misconception: diabetics cannot produce glucagon. They can!!!**
- b. α -cells can respond to fall in blood glucose levels by releasing glucagon. However, there is no stored glycogen to be broken down because diabetics cannot produce insulin to stimulate glycogenesis following a meal.

4. to replace blood glucose taken up;

7. (b) (ii) the blood glucose in the non-diabetic person stayed the same. [2]
1. ref to action of glucagon secreted by α -cells in response to fall in blood glucose levels;
 2. glucagon stimulates liver cells; to break down glycogen to form glucose; which
 3. replace blood glucose taken up / maintain blood glucose concentration at the norm;
- Note: α -cells secrete glucagon – it does not break down glycogen to form glucose.
- (c) Suggest a possible cause of the increase in the blood glucose concentration of the diabetic person at 4.00 pm. [1]
- eaten carbohydrate;;
- (d) Diabetes can be treated by regular injections of a hormone. State **two** sources of that hormone. [2]
1. porcine / bovine insulin;; Accept: insulin from pigs / cattle (Refrain from using “cow”)
 2. genetically engineered bacteria;;

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

8. (a) Describe the behaviour of DNA during the mitotic cell cycle. [8]

Interphase

[max 2]

1. reference to 1 DNA molecule as 1 DNA double helix;
2. DNA exists as decondensed chromatin;
 - a. DNA is coiled / wound around histones; to form nucleosomes;
 - b. DNA is further coiled / folded to form solenoid;
3. reference to euchromatin being decondensed and is thus transcriptionally active;
4. reference to heterochromatin not being fully decondensed and is thus transcriptionally inactive;
5. reference to heterochromatin corresponding to centromeric and telomeric regions;

S phase of interphase

[max 2]

6. DNA molecules undergoes semi-conservative replication;
7. one parental DNA molecule acts as template;
8. for the synthesis of two daughter DNA molecules;
9. which are genetically identical;
10. each daughter DNA molecule is made up of one parental strand and one newly synthesized strand;

Prophase

[max 1]

11. chromatin coil around nonhistone chromosomal proteins; to form looped domains;
12. looped domains coil / fold / condense further compacting all the chromatin to produce discrete chromosomes;
13. each chromosome appears as two identical sister chromatids;
14. joined together at the centromere; by kinetochore proteins;

Metaphase

[max 1]

15. chromosomes are aligned at the metaphase plate;
16. sister chromatids are attached to opposite poles of cell via kinetochore microtubules;

Anaphase

[max 1]

17. sister chromatids of each chromosome separate at centromeric region; and become full-fledged daughter chromosomes;
Reject: split
18. daughter chromosomes move towards opposite poles of the cell; centromere leading;
Reject: opposite ends/ sides

Telophase

19. daughter chromosomes arrive at the poles of the cell;
20. chromosomes decondense into chromatin;

- (b) Using named examples, explain how loss of function mutation and gain of function mutation can contribute to cancer. [8]

Gain of function mutation

[max 4]

- G1. mutation of *ras*; proto-oncogene;
 G2. Ras is a G protein; that relays a signal from a growth factor receptor to a cascade of protein kinases; resulting in stimulation of cell cycle;
 G3. mutation in only one *ras* allele is required to cause the cancer phenotype;
 G4. mutations in the *ras* gene can lead to:
 a. production of a hyperactive Ras protein;
 b. excess amounts of Ras protein;
 G5. kinase cascade is triggered; even in the absence of growth factor, resulting in excessive cell division;

Loss of function mutation

[max 4]

- L1. mutation of *p53*; tumour suppressor gene;
 L2. *p53* protein is a transcription factor; that promotes the expression of genes / synthesis of proteins;
 L3. mutations in both *p53* alleles are required to cause the cancer phenotype;
 L4. mutation of *p53* results in failure to express genes such as
 a. *p21*, hence cell cycle cannot be inhibited / suppressed;
 b. genes, the protein products of which are involved in DNA repair, hence cancer-causing mutations cannot be repaired;
 c. suicide genes, hence apoptosis is inhibited;
 L5. absence of inhibition / suppression; of cell cycle result in uncontrolled cell growth;

- (c) Describe the differences between prokaryotic and eukaryotic mature mRNAs. [4]

Feature	Prokaryotic mRNA	Eukaryotic mRNA
5' capping	modified guanosine triphosphate was <u>not</u> added to the 5' nucleotide;	<u>modified guanosine triphosphate</u> was added to the 5' nucleotide of the pre-mRNA;
3' polyadenylation	adenine nucleotides were <u>not</u> added to the 3' end to create the poly (A) tail;	50 to 250 <u>adenine nucleotides</u> were added to the 3' end to create the <u>poly (A) tail</u> ;
alternative splicing	only one type of mRNA sequence can be <u>derived from each gene</u>	two or more types of mRNA with <u>different / variant sequences</u> can be <u>derived from one gene</u> due to alternative splicing
mono / polycistronic	polycistronic mRNA was transcribed;	monocistronic mRNA was transcribed;

[Total: 20]

9. (a) Explain how the structure and properties of triglycerides are related to their functions in living organisms. [6]

Function		Structure-property
Storage of energy	A1	• <u>large and uncharged</u>; • <u>insoluble</u> in water; • no effect on the <u>water potential</u> of the cells; • prevented from leaving the cells;
	A2	• contain a <u>larger number of hydrogen atoms per gram</u>; • yields a <u>greater amount of energy</u>; • useful for animals in which locomotion requires the mass to be kept to a minimum;
	A3	• oxidised only after carbohydrates are depleted; • long-term energy store → important to hibernating animals;
Source of metabolic water	A4	• 2-fold <u>more hydrogen atoms</u> ; • yield water on <u>oxidation</u> during respiration; • a good source of metabolic water → important to desert animals;
Insulator	A5	• good thermal insulator/poor conductor of heat; • preventing excessive heat loss;
	A6	• electrical insulators; • components of the myelin sheath of nerve cells, allows for rapid transmission of impulse along myelinated neurone;
Buoyancy in animals	A7	• less dense than water; • providing aquatic mammals buoyancy;
	A8	• AVP ;;

- (b) Explain the role of membranes in photosynthesis and respiration. [8]

Photosynthesis

- B1 Structure: thylakoid membrane ;
 B2 Reaction: site of light dependent reactions ;

Respiration

- B3 Structure: inner membrane of mitochondria ;
 B4 Reaction: site of oxidative phosphorylation ;
- B5 to increase surface area ;
 B6 Resp: inner mitochondria membrane highly folded/ form cristae ;
 B7 PS: thylakoid membrane organized into grana ;
 B8 PS: photosynthetic pigments - for maximum absorption of light;
 B9 PS: excite/boost electrons to a higher energy level ;
- B10 impermeable to ions ;
 B11 to allow accumulation of protons/ generation of proton gradient ;
- B12 hold electron carriers of ETC ;
 B13 to allow energy to be released during electron transfer down ETC ;
 B14 to harness energy released to pump H⁺ ;
- B15 hold stalked particles / ATP synthase ;
 B16 ATP synthase couples the movement of H⁺ down its gradient ;
 B17 to the phosphorylation of ADP to ATP ;

- (c) Compare haemoglobin with insulin in term of structure and function. [6]

Similarities

- C1 Globular ;;
 C2 hydrophobic amino acids are buried in the interior of the globular structure whilst the hydrophilic amino acid residues are on the outside ;;
 C3 Structure maintained by disulfide bonds, ionic bonds, hydrogen bonds and hydrophobic interactions ;;

Differences

		Haemoglobin	Insulin
Structure	C4	Made up of <u>4</u> polypeptide chains: 2 α subunits, 2 β subunits;	Made up of <u>1 (or 2)</u> polypeptide chain;
	C5	<u>Quaternary</u> structure;	<u>Tertiary</u> structure ;
	C6	Consists of <u>non-protein</u> component: haem group;	Consists of protein component only;
Function	C7	Transport oxygen/ respiratory pigment;	Regulate glucose metabolism/ hormone;

[Total: 20]